# Constellations and formation flying

# 6

*Danil Ivanov and Mikhail Ovchinnikov*
Space Systems Dynamics Department, Keldysh Institute of Applied Mathematics of Russian Academy of Sciences, Moscow, Russia

## 1 Distributed space system definitions and features

The increasing complexity of approaches and methods of solving fundamental and applied problems of space exploration often leads to the need to create distributed space systems (DSS). In this case, several satellites united in one system and flying at a relatively short distance from each other or at similar orbits are simultaneously working on solving a common problem. The use of small satellites in such projects is the most natural and often the only possible way.

According to the most common definition, DSS is a system consisting of multiple space elements that can communicate, coordinate, and interact to achieve a common goal. Its main features are tolerance for failure of individual satellites, scalability and flexibility in design and deployment, and new capabilities compared with a single satellite. There are two main DSS types that can be distinguished depending on the degree of autonomy, communication between the satellites, number of satellites, and types of relative trajectories. These main DSS types are satellite constellations and satellite formations. Satellite constellations are characterized by similar orbits, the absence of intersatellite communication, and the lack of control of relative motion, and as a rule, the individual satellites are controlled from the ground. Satellite formations are associated with small relative intersatellite distances compared with the orbital size, onboard closed-loop control to preserve relative motion topology and often are characterized by supporting interspacecraft communication between the individual satellites. With the development of nano- and picosatellites (such as CubeSats and ChipSats) and the possibility of their standardized and mass production, a new class of formation flying began to develop—the so-called swarm of satellites. The swarm consists of a large number of near-flying satellites moving along arbitrary but limited relative trajectories.

Despite clearly defined features of types of distributed space systems, sometimes disputes arise as to whether a particular system belongs to a particular category. For example, satellites challenge the definitions in case they move along close relative trajectories but they are controlled from the ground or in case satellites have communication links but lack relative motion control. In such cases the degree of autonomy is the most crucial aspect. On-ground motion control and limited autonomy define these architectures as constellations. There are two main approaches to the autonomous

**CubeSat Handbook.** https://doi.org/10.1016/B978-0-12-817884-3.00006-0

control of a group of satellites: centralized control and decentralized control. Centralized control implies the presence of a "head" or "mother" satellite in the formation; its motion is monitored by the remaining "deputy" or "daughter" satellites that are controlled to achieve the required relative trajectories. Centralized control is more suitable for small groups of satellites moving along predetermined relative trajectories. For a significant number of satellites like the swarm, this approach seems to be not reasonable since the head satellite may be beyond the communication range for some satellites. Additionally, intersatellite communication always has a limited number of channels. This makes it difficult to determine the motion of the "mother" satellites relative to the "daughters." With the decentralized control approach, each satellite makes the decision to control its dynamic states, modes, and activities individually based on the information on motion of the nearest neighbors. This approach is more suitable for a swarm of satellites taking into account the communicational limitations.

## 2 CubeSats constellations: Control problems and solutions

Constellations of CubeSats firmly occupy a place among well-known constellations of satellites of greater mass. The low cost of CubeSat production and launch allows placing significant number of satellites on orbit. Though the reliability and obtained data quality still leave much to be desired for CubeSat constellations, the processing of the large amount of measurements outweighs all the disadvantages of the capabilities of this type of orbital system.

The main parameter of the constellation is coverage of the region of interest on the Earth surface or in near-Earth space, which depends on the mission goal. In the case that the coverage is not continuous, the period of repeatability is also of great importance. These constellation factors directly depend on the selected satellite orbits and number of satellites in each orbital plane. However, the orbits degrade over time due to perturbations, so to minimize natural relative changes, as a rule similar orbits for satellites are considered for constellations. For example, if the inclinations of the orbits are the same, then the drift in the right ascension of ascending node (RAAN) is similar due to $J_2$ influence, and the relative orbit positions are maintained. A popular near-circular orbit geometry is the Walker Delta Pattern constellation, where a number of satellites are evenly distributed along the set of equally spaced orbital planes with the same inclination [1]. To construct this type of constellation, it is necessary to place each satellite in a specific location in orbit. There are two ways to do this: (1) to use launcher capabilities or (2) to perform maneuvers using onboard propulsion after the launch. In the case of the CubeSat constellation, often a cluster launch is applied when all the satellites are to be launched in a short time as a secondary payload in nearly one location in orbit. After such a cluster deployment, the satellites need to separate along the orbit. However, the challenge is that the propulsion systems onboard CubeSats are typically limited in capability or simply absent. The ability of CubeSats to maneuver from their insertion orbit into their required mission orbits is therefore restricted. This is especially true if expensive out-of-plane maneuvers in RAAN or inclination are required.

To achieve mission-required orbits, a set of constellation deployment strategies have been developed and implemented, which allows the mission team to distribute satellites that are launched together into the working orbit positions [2]. For orbital plane separation, a differential rate of nodal precession is used. Due to the non-spherical geopotential of Earth orbits with different sizes, shapes, or orientation, the orbits precess at different rates, allowing plane separations to be achieved without out-of-plane maneuvering. The idea of this deployment method is to perform an in-orbit maneuver for one of the satellite to obtain an orbit with different nodal precession rate. After a time, a natural drift will lead a required orbital plane separation in RAAN, and then the second satellite performs a maneuver to achieve the orbit with the same nodal precession rate to fix the difference in RAAN. This procedure can be repeated to obtain all required orbital planes separation for a constellation. In the case of CubeSats without propulsion, the maneuvers for changing the relative nodal drift rate can be executed by carrier vehicle that launches the satellites into the orbits with required orbital plane separation [3]. This method of constellation deployment using differential nodal drift rates has been demonstrated by the FORMOSAT/COSMIC mission launched in 2006 [4].

Another problem of constellation deployment in the case of cluster launch is the phasing of the satellites along the orbit. It can be formulated as a problem of changing the satellite phase—an angle describing the position of satellites along its orbit. For constellations, the satellites in one orbital plane are usually required to be uniformly distributed in relative phase or just to be at the specified relative phases. The required maneuvers using onboard propulsion for solution of this problem are well studied [5–7] and have been implemented for massive satellite constellations. For CubeSats without propulsion, the problem can be solved using differential atmospheric drag in the case of LEO orbits. In this case the satellites are able to vary their individual cross-sectional areas with respect to the incoming airflow; it is possible to achieve the difference in the drag force acting on the satellites. This natural force is applied to control the relative orbital phase between the satellites in the constellation. This control scheme for solving phasing problem was first implemented in 2013 during the AeroCube-4 mission. AeroCube-4 consisted of three 1U CubeSats with deployable solar panels [8]. Later, this methodology was successfully used by Planet Inc. for dispersing their 3U CubeSats called Doves along a single orbital plane [9]. The record number of 88 Doves in one flock was launched in 2017 and was separated uniformly along the orbit in 8 months [10]. By mid-2019 the fleet of planet's constellation had 170 active CubeSats, making it the largest constellation to date [11].

CubeSat constellations that have been launched and that are planned are listed on the web page "Nanosat Database" [12], which lists the most up-to-date information on the number of CubeSats in constellations, their form factor, and also the field of their application. The second largest commercial constellation by number of satellites and the largest by number of sensors is Spire Global fleet, consisting of 70 3U CubeSats called Lemur [13]. Other CubeSats constellations are in the initial stages of deployment, and their number does not exceed seven satellites to date.

**Table 1** Summary of CubeSat constellation deployment problems in case of cluster launch.

| Control problem | Solution | Onboard hardware required |
|---|---|---|
| Distribution of satellites along orbital planes with separation in RAAN | Perform the in-plane maneuvers and use natural drift to obtain required orbital plane separation in RAAN | Propulsion, three-axis attitude control system |
| Phasing satellites along a single orbital plane | Differential aerodynamic drag application in case of the launch in LEO | Active attitude control system, high area-to-mass ratio form factor or drag sail |

A summary of the Section 2 is presented in Table 1.

# 3 Nanosatellites formation flight control

According to the equations of relative free motion in the common case of two satellites flying at close distances, the satellites will move along in an unbounded elliptical spiral relative trajectory. That is why the relative motion control is needed for keeping close formation flight. The control in formation flying can be used for various tasks: tracking or maintenance of the required relative motion, reconfiguration, proximity operations, and even docking to each other. This section describes the main features of CubeSat formation flying control, paying particular attention to their modest technical abilities.

## *3.1 Relative navigation problems*

Autonomous relative motion determination is one of the main problems of satellite formation flying. Formation flying control algorithms utilize the relative state estimation obtained using processing of measurements based on relative motion. In this section the main approaches to relative navigation are considered along with emphasis on the applicability to CubeSats.

The measurements that can be directly obtained include physical parameters such as relative distances determined through radio frequency or laser ranging. However, lasers require high attitude pointing accuracy, which is a challenge for low-cost CubeSat missions. Another approach is to use GPS receivers onboard the satellites. The absolute position of each satellite in Earth-fixed reference frame is transmitted to the other satellites via intersatellite communication links; thus the difference of absolute position vectors provides the relative position. In the CanX-4&5 missions launched in 2014, this strategy using relative navigation based on GPS receivers was implemented [14, 15]. The experimentally obtained relative position estimation errors were about 10 m.

Image processing is often used for relative state determination in formation flying. An observing camera is required to be installed to take pictures of the other satellites in the group. Depending on the camera parameters and relative position, it is possible to estimate not only translational motion but also rotational motion. Satellite images with so-called feature points are recognized, defining known positions in the spacecraft-related reference frame. Using these measurements of feature point positions in the imagery and projective geometry equations, the relative position and orientation is estimated. A paper published by Opromolla et al. [16] provides an overview of the existing visual-based systems and the state of the art of the cooperative and uncooperative satellite pose determination techniques. A set of works are devoted to the developing and flight testing of the algorithms for relative position determination of the unknown target using angle-only visual navigation [17–19]. The relative navigation system based on image processing was implemented in a set of formation flying missions. Among them the PRISMA mission tested a wide variety of vision-based navigation approaches [20]. Though vision-based navigation is the most reliable, image processing in real time with available onboard CubeSat computers that typically have low computational power represents a challenge. Also by means of economy, the miniature star tracker can be used also as a navigation camera.

## 3.2 Restricted propulsion and control approaches without propellant

The conventional approach to formation flying is to use onboard propulsion for producing the required force to control relative position in the formation. To be able to implement a given thrust direction, a three-axis attitude control system must also be installed onboard. Such systems with full controllability are often used on satellites of a large size and mass, and a large number of different control algorithms have been developed for these large-scale missions [21–23]. As has been previously noted, the CubeSat ability to perform any maneuvers is restricted due to mass, cost, and volume constraints. There could be no propulsion system onboard at all. That is why in this subsection, we will focus on control approaches with restricted propulsion systems and fuelless control approaches.

In the case that the satellites are equipped with propulsion systems but there is no three-axis attitude stabilization onboard, the so-called single-input control approach has been developed. It is assumed that the thrust vector is fixed with respect to the body reference frame, and a number of simple and lightweight attitude control systems are available that can stabilize motion of that thrust axis. For example, if the satellite is equipped with a passive magnetic attitude control system that allows it to stabilize the longitudinal axis of the satellite with permanent magnet along the local geomagnetic field, a single-input control is able to achieve bounded relative trajectories of the two satellites in near-circular orbit [24]. Also the satellite can be stabilized by rotation along the rotational axis with a minimal inertia moment. In a 2012 paper [25], Guerman et al. show that it is possible to obtain the closed relative trajectories taking into account $J_2$ perturbation in the cases of spin stabilization and passive magnetic stabilization, though the shape of the closed relative trajectories depends on the orbital parameters and initial conditions.

Nowadays a number of nonconventional approaches for formation flying control have been proposed. The general idea that is the basis of these approaches is to develop an effective control method that does not require fuel consumption. Environmental forces, such as aerodynamic drag (AD) force and solar radiation pressure (SRP) force, can be used. Both approaches require a sail or form factor of satellites with high area-to-mass ratio. The CubeSats of 3U size are already appropriate for differential drag application. The principal idea here is to use a difference in environmental forces acting on each satellite in formation flying. This difference usually appears when the satellite changes its relative attitude but the effective size variation is also considered in the literature. Though in general the AD and SRP acceleration models are similar, there are a number of differences. One can apply AD control in low Earth orbits (LEO) only, and its value is varying due to the atmosphere density change caused by the day/night variations and the orbital altitude variations. So, it is usually difficult to predict the exact value of the control force. SRP can be used in different types of orbits, but the shaded parts of orbit should be taken into consideration.

Control based on the aerodynamic drag force was first proposed for the formation flying in 1986 by Leonard [26] under the assumption of a discrete change in the effective cross section of satellites flying in the group. He developed a control algorithm based on the proportional differential controller. A large number of papers applied a variety of the different control algorithms using differential drag including PID regulators [27], linear-quadratic regulators [28], Lyapunov-based control [29, 30], sliding mode control [31], and optimal control [32]. These papers only considered the aerodynamic differential drag and allow control of the relative motion in a single orbital plane only. Some recent papers take into account the differential lift and also consider the spatial relative motion control. The application of the differential lift along with the drag for the small satellite rendezvous problem was first proposed by Horsley et al. [33]. Control strategy developed in Ref. [33] is based on the bang-bang approach when only the maximum values of the lift and drag are used. Papers [28, 34] address the problem of the satellite formation keeping by the differential lift and drag under the $J_2$ perturbation. The application AD for construction of tetrahedral formation flying using 3U CubeSats is considered in a paper by Ivanov et al. [35].

The idea of SRP application for CubeSat formation flying control is inspired by successful single satellite missions like LightSail 2 and NanoSail-D2 with sails onboard. The CubeSats are equipped with active attitude control systems to be able to orient the normal vector to the sail relative to the Sun direction to produce required SPR force. A set of papers are devoted to the control algorithm using conventional sail [36–41] and sails with variable reflectivity properties [42].

The classical control approach using propulsion and the use of SRP or AD do not affect the relative motion directly, but control the orbital motion of one or several satellites instead. This seems ineffective because the significant amount of acceleration is aimed to the whole formation flying momentum change. The electrostatic and the electromagnetic control approaches are free from these shortcomings. Since both forces appear when two or more satellites in a formation have either electrostatic charge or dipole moment, the total momentum of the interacting satellites conserves. So, these approaches allow controlling relative motion directly.

A concept of electrostatic formation flying control was suggested by King et al. in 2002 [43]. This concept is based on the results of the SCATHA mission where the satellite electrostatic charge system was first tested. The miniature cold-cathode electron gun for the charging was used in Aalto-1 and in the ESTCube-1 3U CubeSat missions. The onboard gun charged deployed tether for the application of the Lorentz force in the geomagnetic field for faster deorbiting. Changing the value of this force can be used for formation flying control as well.

The concept of tethered satellite formation flying can also be implemented using CubeSats. The main idea is to link two or more satellites by some rope and to control the relative motion by changing its length [44–47]. The realization of this concept is complicated because of flexible rope motion [48]. So the relative motion of the two connected by tether satellites is usually designed to provide the tension of the tether.

A summary of the Section 3.2 is presented in Table 2.

## 3.3 Swarm of nanosatellites decentralized control

Swarm-type flight requires only bounded relative motion with no other restrictions in contrast to the formation type of constellation. Random relative trajectories in the swarm have several advantages: (1) economy of the control source of the satellites, (2) less dependence on the failure of the specific satellite, and (3) reduced demands for the onboard hardware and software. During the launch of the swarm, some error in the separation speed of nanosatellites from the launch vehicle is inevitable. This leads

**Table 2** Summary of CubeSat formation flying control approaches and its features.

| Control approach | Motion features | Onboard hardware required |
|---|---|---|
| Single-input control | Able to achieve the closed relative trajectories, though its form and shape depends on orbital parameters and initial conditions | Propulsion, passive attitude control system |
| Differential AD | Allows to control the relative motion mainly in the orbital plane. Applicable for LEO | Active attitude control system, high area-to-mass ratio form factor or drag sail |
| Differential SRP | Applicable to orbits without Earth shading | Active attitude control system, solar sail |
| Lorentz force | Applicable for LEO and MEO | Electron gun system |
| Tether | The size of the relative orbit is limited by the tether length. Relative motion is chosen so as to achieve tether tension | Tether deployment system |

to a slightly different orbital period of the satellites, the relative trajectories become unlimited, and the swarm degrades. Thus the problem of the relative drift reduction or elimination is of great importance.

For control of each satellite in the swarm, it is necessary that the information about the relative motion of all other satellites be made available. However, in the case of a significant number of satellites in the swarm, this is a difficult task due to the hardware limitations of the relative motion determination system and/or the intersatellite communication limitations. These restrictions that prevent the availability of information about the relative motion of all satellites in the swarm will henceforth be referred to as communicational constraints. In natural swarms, for example, for a swarm of insects, there is a limit to the number of communication links between each element of the swarm and its neighbors. In addition, there is a limit of the maximum distance between the elements of the swarm at which the relative motion can be known. Similar restrictions exist for a swarm of satellites. Each satellite can estimate the relative motion of other satellites using onboard motion determination systems. However, due to the limited capabilities of the sensors, each satellite can estimate the relative motion of those satellites that are located in a certain neighborhood only. Relative motion determination systems can be based on the processing of images, range finders, and other sensors, but the number of satellites whose relative motion can be determined simultaneously is generally small. Also, there is a limit on the range at which the system can operate with the necessary accuracy. These features of the autonomous motion determination system can be overcome by sharing information between satellites about the current orbital motion obtained, for example, using onboard GPS receivers installed on each satellite, and thus the relative motion can be computed [14]. However, intersatellite communication channels also cannot provide unlimited number of connections to one satellite owing to frequency restrictions of signals. Thus, during deployment of a swarm of satellites, it is necessary to take into account the features associated with communication constraints as described in a study by Ivanov et al. [49] in which they considered a set of control strategies for 20 3U CubeSats swarm deployment.

The so-called "agents" are considered in the literature about swarm motion to be independent and autonomous controllable units, in our case the satellites. In most papers on multiagent systems, a control plays according to four rules: attraction, alignment, collision avoidance, and achievement of the goal. In a study by Sabatini et al. [50], a linear-quadratic regulator using these rules to control the swarm of satellites is considered, and a comparison of characteristic velocity of centralized and decentralized strategies is performed for various parameters and tasks of the mission. A reduction in computational complexity for a large number of satellites is shown when a two-stage control (a combination of centralized and decentralized strategies) is used. A follow-on paper by the same authors [51] focuses on the study of the decentralized approach using an artificial potential function for control; however, the dynamics of the relative motion of satellites formation flying is not considered.

A propulsion system can be used for the swarm deployment, but the differential drag-based control is more suitable for nanosatellites in LEO. It does not require propellant; however, the active attitude control system is needed. However, almost all the studies on differential drag control consider only two satellites in formation flying with application of the centralized control approach [26–32]. A few papers are devoted to differential drag control of the multiple satellites. The cyclic and optimal control strategies for a cluster flight with more than two satellites are proposed in a paper by Ben-Yaacov and Gurfil [52]. Stability and performance of cluster keeping while avoiding collisions were studied in 2015 by the same researchers [53]. The papers mentioned earlier do not address communication restrictions and decentralized control features.

The most critical issue in swarm control is collision avoidance between satellites moving along random relative trajectories. The collision avoidance algorithms are usually aimed at decreasing the probability of a dangerous rendezvous between the satellites. So the algorithm described in the work by Slater et al. [54] is intended to calculate the necessary maneuver for a predicted close distance and to minimize the collision probability and characteristic velocity. In work by Bombardelli and Hernando-Ayuso [55], an optimum algorithm is proposed to minimize the fuel consumption and maximize the relative distance between satellites in the same time. Another common approach for preventing collisions is the method of artificial potential fields considered [56]. The satellite forms a spherical potential field around itself. According to the control algorithm, if another satellite enters this sphere, a repulsive force directed along the radius from one satellite to the second acts on it. Schlanbusch et al. [57] describes an algorithm based on behavioral control for the reconfiguration of a group of satellites. If the satellites enter a spherical forbidden zone during reconfiguration, the satellite is affected by a predetermined repulsive pulse directed along the radius vector. Such an impulse is applied as long as the satellite is in the forbidden zone.

The 3U CubeSat KickSat-2 deployed 105 so-called ChipSats or Sprites in March 2019 representing the first launched swarm mission so far. The Sprites are 3.5-cm by 3.5-cm chips with all the satellite systems necessary for self-contained operations [58]. Sprite-like satellites could be distributed by the hundreds or thousands in orbit and could become a spatially distributed measurement system that even could form a planetary ring [59]. The ChipSats launched to date have no relative motion control, and they are flying apart passively, but small magnetic control system can be installed and differential drag applied in future missions. Another idea by the StartRocket company is to launch a swarm of CubeSats with sunlight reflectors for constructing the orbital display. Each satellite will represent a pixel in some image that can be seen from the Earth under appropriate illumination conditions [60]. The control of such a display is planned to be implemented using either onboard thrusters or using differential aerodynamic forces. This and the other recent developments in attitude control described earlier will likely facilitate the advancement of SmallSat constellations, formations, and swarms with an extremely wide variety of applications in Earth remote sensing, space weather monitoring, and other applications yet to be explored.

# References


[1] J.G. Walker, Satellite constellations, J. Br. Interplanet. Soc. 37 (1984) 559–571.

[2] N.H. Crisp, K. Smith, P. Hollingsworth, Launch and deployment of distributed small satellite systems, Acta Astronaut. 114 (2015) 65–78, https://doi.org/10.1016/j.actaastro.2015.04.015.

[3] Method and Apparatus for Deploying a Satellite Network, https://patents.google.com/patent/US5199672A/en, 1990. (Accessed 29 May 2019).

[4] C.-J. Fong, W.-T. Shiau, C.-T. Lin, T.-C. Kuo, C.-H. Chu, S.-K. Yang, N.L. Yen, S.-S. Chen, Y.-H. Kuo, Y.-A. Liou, S. Chi, Constellation deployment for the FORMOSAT-3/COSMIC mission, IEEE Trans. Geosci. Remote Sens. 46 (2008) 3367–3379, https://doi.org/10.1109/TGRS.2008.2005202.

[5] S.P. Trofimov, M.Y. Ovchinnikov, Optimal multiple-impulse circular orbit phasing, J. Guid. Control. Dyn. 39 (2016) 1678–1681, https://doi.org/10.2514/1.G001513.

[6] C.D. Hall, V. Collazo-Perez, Minimum-time orbital phasing maneuvers, J. Guid. Control. Dyn. 26 (2003) 934–941, https://doi.org/10.2514/2.6921.

[7] C. Cappelletti, S. Battistini, F. Graziani, Small launch platforms for micro-satellites, Adv. Space Res. 62 (2018) 3298–3304, https://doi.org/10.1016/J.ASR.2018.05.004.

[8] J. Gangestad, B. Hardy, D. Hinkley, Operations, orbit determination, and formation control of the AeroCube-4 CubeSats, in: 27th Annu. AIAA/USU Conf. Small Satell. No. SSC13-X-4, Logan, Utah, 2013.

[9] C. Boshuizen, J. Mason, P. Klupar, S. Spanhake, Results from the Planet Labs Flock Constellation, in: AIAA/USU Conf. Small Satellites, 2014. https://digitalcommons.usu.edu/smallsat/2014/PrivEnd/1. (Accessed 29 May 2019).

[10] C. Foster, J. Mason, V. Beukelaers, L. Stepan, R. Zimmerman, Differential drag control scheme for large constellation of planet satellites and on-orbit results, in: Proceedings Int. Work. Satell. Constellations Form. Flying, June 19-21, 2017, Boulder, CO, 2017, pp. 1–18.

[11] Planet, Daily Satellite Imagery and Insights, n.d. https://www.planet.com/ (Accessed 29 May 2019).

[12] Nanosats Database, n.d. https://www.nanosats.eu/tables#constellations (Accessed 29 May 2019).

[13] Spire, Space to Cloud Data & Analytics, n.d. https://www.spire.com/en (Accessed 29 May 2019).

[14] N.H. Roth, B. Risi, R.E. Zee, Flight results from the Canx-4 and Canx-5 formation flying mission, in: Proc. Int. Work. Satell. Constellations Form. Flying, June 19-21, 2017, Boulder, CO, 2017, pp. 1–17.

[15] G. Bonin, N. Roth, S. Armitage, J. Newman, B. Risi, R.E. Zee, CanX-4 and CanX-5 precision formation flight: mission accomplished! in: 29th Annu. AIAA/USU Conf. Small Satell. Uta, LoganSSC15-I-4, 2015. https://pdfs.semanticscholar.org/c5bd/33760a1b8c80781d299999c4ebe55ebc29da.pdf. (Accessed 29 May 2019).

[16] R. Opromolla, G. Fasano, G. Rufino, M. Grassi, A review of cooperative and uncooperative spacecraft pose determination techniques for close-proximity operations, Prog. Aerosp. Sci. 93 (2017) 53–72, https://doi.org/10.1016/J.PAEROSCI.2017.07.001.

[17] F. Perez, D. Modenini, A. Vázquez, F. Aguado, R. Tubío, G. Dolgos, A. Gonzalez, R. Lasagni Manghi, M. Zannoni, A. Nazeeruddin, M. Melozzi, I. Carnelli, DustCube, a nanosatellite mission to binary asteroid 65803 Didymos as part of the ESA AIM mission, Adv. Space Res. 62 (2018) 3335–3356, https://doi.org/10.1016/J.ASR.2018.06.019.

[18] S. D'Amico, J.-S. Ardaens, G. Gaias, H. Benninghoff, B. Schlepp, J.L. Jørgensen, Non-cooperative rendezvous using angles-only optical navigation: system design and flight results, J. Guid. Control. Dyn. 36 (2013) 1576–1595, https://doi.org/10.2514/1.59236.

[19] D. Ivanov, M. Ovchinnikov, M. Sakovich, Relative pose and inertia determination of unknown satellite using monocular vision, Int. J. Aerosp. Eng. (2018) 1–16, https://doi.org/10.1155/2018/9731512.

[20] R. Noteborn, Flight results from the PRISMA optical line of sight based autonomous rendezvous experiment, in: Proc. 4th Int. Conf. Spacecr. Form. Fly. Mission. an Technol. May 2011, Montr. Canada, 2011, p. 10.

[21] K. Alfriend, S. Vadali, P. Gurfil, J. How, L. Breger, Spacecraft Formation Flying: Dynamics, Control and Navigation, Elsevier/Butterworth-Heinemann, Oxford, 2010.

[22] D.P. Scharf, F.Y. Hadaegh, S.R. Ploen, A survey of spacecraft formation flying guidance and control (part 1): guidance, in: Proc. 2003 Am. Control Conf, IEEE, 2003, pp. 1733–1739, https://doi.org/10.1109/ACC.2003.1239845.
[23] G. Di Mauro, M. Lawn, R. Bevilacqua, Survey on guidance navigation and control requirements for spacecraft formation-flying missions, J. Guid. Control. Dyn. 41 (2018) 581–602, https://doi.org/10.2514/1.G002868.
[24] M.Y. Ovchinnikov, D. Bindel, D.S. Ivanov, G.V. Smirnov, S. Theil, I.E. Zaramenskikh, Development and laboratory verification of control algorithms for formation flying configuration with a single-input control, Acta Astronaut. 67 (2010) 1158–1163.
[25] A.D. Guerman, M.Y. Ovchinnikov, G.V. Smirnov, S.P. Trofimov, Close relative trajectories for formation flying with single-input control, Math. Probl. Eng. 2012 (2012) 20, https://doi.org/10.1155/2012/967248 Article ID 967248.
[26] C.L. Leonard, Formation Keeping of Spacecraft Via Differential Drag (Master thesis). Massachusetts Inst. Technol, Cambridge, MA, 1986.
[27] B.S. Kumar, A. Ng, A. Bang-Bang, Control approach to maneuver spacecraft in a formation with differential drag, in: Proc. AIAA Guid. Navig. Control Conf. Exhib. AIAA Pap. No. 2008–6469, Honolulu, Hawaii, August, 2008.
[28] D. Ivanov, M. Kushniruk, M. Ovchinnikov, Study of satellite formation flying control using differential lift and drag, Acta Astronaut. 145 (2018) 88–100, https://doi.org/10.1016/J.ACTAASTRO.2018.07.047.
[29] D. Pérez, R. Bevilacqua, Lyapunov-based adaptive feedback for spacecraft planar relative maneuvering via differential drag, J. Guid. Control. Dyn. 37 (2014) 1678–1684, https://doi.org/10.2514/1.G000191.
[30] D. Pérez, R. Bevilacqua, Differential drag spacecraft rendezvous using an adaptive Lyapunov control strategy, Acta Astronaut. 83 (2013) 196–207, https://doi.org/10.1016/j.actaastro.2012.09.005.
[31] K.D. Kumar, A.K. Misra, S. Varma, F. Bellefeuille, Maintenance of satellite formations using environmental forces, Acta Astronaut. 102 (2014) 341–354, https://doi.org/10.1016/j.actaastro.2014.05.001.
[32] L. Dellelce, G. Kerschen, Optimal propellantless rendez-vous using differential drag, Acta Astronaut. 109 (2015) 112–123, https://doi.org/10.1016/j.actaastro.2015.01.011.
[33] M. Horsley, S. Nikolaev, A. Pertica, Small satellite rendezvous using differential lift and drag, J. Guid. Control. Dyn. 36 (2013) 445–453, https://doi.org/10.2514/1.57327.
[34] X. Shao, M. Song, J. Wang, D. Zhang, J. Chen, Satellite formation keeping using differential lift and drag under J2 perturbation, Aircr. Eng. Aerosp. Technol. 89 (2017) 11–19.
[35] D. Ivanov, M. Mogilevsky, U. Monakhova, M. Ovchinnikov, A. Chernyshov, Deployment and maintenance of nanosatellite tetrahedral formation using aerodynamic forces, in: Proc. IAC-2018. Pap. IAC-18-B4.7.6, 2018, p. 11.
[36] S. Gong, G. Yunfeng, J. Li, Solar sail formation flying on an inclined Earth orbit, Acta Astronaut. 68 (2011) 226–239, https://doi.org/10.1016/j.actaastro.2010.08.022.
[37] K. Shahid, K.D. Kumar, Multiple spacecraft formation reconfiguration using solar radiation pressure, Acta Astronaut. 103 (2014) 269–281, https://doi.org/10.1016/j.actaastro.2014.05.021.
[38] K. Shahid, K.D. Kumar, Formation control at the Sun-Earth L(2) libration point using solar radiation pressure, J. Spacecr. Rocket. 47 (2010) 614–626, https://doi.org/10.2514/1.47342.
[39] S. Gong, H. Baoyin, J. Li, Solar sail formation flying around displaced solar orbits, J. Guid. Control. Dyn. 30 (2007) 1148–1152, https://doi.org/10.2514/1.24315.
[40] T. Williams, Z.-S. Wang, Solat radiation pressure and formation-keeping in highly elliptical orbits, in: AIAA/AAS Astrodyn. Spec. Conf. Exhib, 2002, pp. 1–14.

[41] K. Parsay, H. Schaub, Drift-free solar sail formations in elliptical Sun-synchronous orbits, Acta Astronaut. 139 (2017) 201–212, https://doi.org/10.1016/J.ACTAASTRO.2017.06.027.
[42] Y.V. Mashtakov, M.Y. Ovchinnikov, T.Y. Petrova, S.S. Tkachev, Attitude and relative motion control of satellites in formation flying via solar sail with variable reflectivity properties, in: Proc. 69th International Astronaut. Congr, 2018, p. 8.
[43] L.B. King, G.G. Parker, S. Deshmukh, J.-H. Chong, Spacecraft Formation-Flying using Inter-Vehicle Coulomb Forces, Tech. Rep. NASA/NIAC (2002) 103 p.
[44] S.-J. Chung, D.W. Miller, Propellant-free control of tethered formation flight, part 1: linear control and experimentation, J. Guid. Control. Dyn. 31 (2008) 571–584, https://doi.org/10.2514/1.32188.
[45] J. Zhang, K. Yang, R. Qi, Dynamics and offset control of tethered space-tug system, Acta Astronaut. 142 (2018) 232–252, https://doi.org/10.1016/J.ACTAASTRO.2017.10.020.
[46] Y. Chen, R. Huang, X. Ren, L. He, Y. He, History of the tether concept and tether missions: a review, ISRN Astron. Astrophys. 2013 (2013) 1–7, https://doi.org/10.1155/2013/502973.
[47] A.D. Guerman, G. Smirnov, P. Paglione, A.M. Vale Seabra, Stationary configurations of a tetrahedral tethered satellite formation, J. Guid. Control. Dyn. 31 (2008) 424–428, https://doi.org/10.2514/1.31979.
[48] S.S. Gates, S.M. Koss, M.F. Zedd, Advanced tether experiment deployment failure, J. Spacecr. Rocket. 38 (2001) 60–68.
[49] D. Ivanov, U. Monakhova, M. Ovchinnikov, Nanosatellites swarm deployment using decentralized differential drag-based control with communicational constraints, Acta Astronaut. 159 (2019) 646–657, https://doi.org/10.1016/J.ACTAASTRO.2019.02.006.
[50] M. Sabatini, F. Reali, G.B. Palmerini, Autonomous behavioral strategy and optimal centralized guidance for on-orbit self assembly, in: IEEE Aerosp. Conf. Proc., 12, 2009, https://doi.org/10.1109/AERO.2009.4839582.
[51] M. Sabatini, G.B. Palmerini, P. Gasbarri, Control laws for defective swarming systems, Adv. Astronaut. Sci. 153 (2015) 749–768.
[52] O. Ben-Yaacov, P. Gurfil, Long-term cluster flight of multiple satellites using differential drag, J. Guid. Control. Dyn. 36 (2013) 1731–1740, https://doi.org/10.2514/1.61496.
[53] O. Ben-Yaacov, P. Gurfil, Orbital elements feedback for cluster keeping using differential drag, Adv. Astronaut. Sci. 153 (2015) 769–787, https://doi.org/10.1007/s40295-014-0022-0.
[54] G.L. Slater, S.M. Byram, T.W. Williams, Collision avoidance for satellites in formation flight, J. Guid. Control. Dyn. 29 (2006) 1140–1146, https://doi.org/10.2514/1.16812.
[55] C. Bombardelli, J. Hernando-Ayuso, Optimal impulsive collision avoidance in low Earth orbit, J. Guid. Control. Dyn. 38 (2015) 217–225, https://doi.org/10.2514/1.G000742.
[56] D. Lee, A.K. Sanyal, E.A. Butcher, Asymptotic tracking control for spacecraft formation flying with decentralized collision avoidance, J. Guid. Control. Dyn. 38 (2015) 587–600, https://doi.org/10.2514/1.G000101.
[57] R. Schlanbusch, R. Kristiansen, P.J. Nicklasson, Spacecraft formation reconfiguration with collision avoidance, Automatica 47 (2011) 1443–1449, https://doi.org/10.1016/j.automatica.2011.02.014.
[58] KickSat, n.d. https://kicksat.github.io/ (Accessed 3 September 2019).
[59] C. Colombo, C. McInnes, Orbit design for future SpaceChip swarm missions in a planetary atmosphere, Acta Astronaut. 75 (2012) 25–41, https://doi.org/10.1016/J.ACTAASTRO.2012.01.004.
[60] The Orbital Display by StartRocket, n.d. https://startrocket.me/ (Accessed 3 September 2019).